

Browsing and Searching Compressed Documents

Raymond Wan

Department of Computer Science and Software Engineering
The University of Melbourne

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Abstract

Compression and information retrieval are two areas of document management that exist separately due to the conflicting methods of achieving their goals. This research examines a mechanism which provides lossless compression and phrase-based browsing and searching of large document collections. The framework for the investigation is an existing off-line dictionary-based compression algorithm.

An analysis of the algorithm, supported by previous work and experiments, highlights two factors that are important for retrieval: efficient decoding, and a separate dictionary stream. However, three areas of improvement are necessary, prior to the inclusion of the algorithm into a browsing system.

First, in order to accommodate retrieval, the algorithm must produce a dictionary built up on words, rather than characters. A pre-processing stage is introduced which separates the message into words and non-words, along with word modifiers.

Second, the memory requirements of the algorithm prevent the processing of large documents. Earlier work has proposed a solution which separates the message into individual blocks prior to compression. Here, a post-processing stage is proposed which combines the blocks in a series of phases. Experiments show the trade-offs between the number of phases performed and the improvements in compression levels.

Organisations which provide access to an information retrieval system to users are sometimes concerned with the amount of disk space available. But users have a different point of view and may place response time at a higher priority. Indeed, faster computers and network connections translate into plummeting patience levels of users. The last improvement to the compression algorithm is two new coding schemes which replace the entropy coder that was used in previous work. While deploying them sacrifices compression effectiveness, these two mechanisms offer improved efficiency, as shown through experiments.

With the enhancements to the compression algorithm in hand, a technique to efficiently support phrase browsing is presented. Phrase contexts can be searched and progressively refined through the word modifiers. Because of the three changes to the algorithm, phrases are more visually appealing, larger documents can be processed, and response times are improved.

Declaration

This is to certify that

1. the thesis comprises only my original work towards the PhD except where indicates in the Preface,
2. due acknowledgement has been made in the text to all other material used, and
3. the thesis is less than 100,000 words in length, exclusive of tables, maps, bibliographies, and appendices.

Raymond Wan

BSc (University of British Columbia, 1997)

Preface

Publications arising from this thesis.

The general concept of phrase browsing documents compressed with character-based RE-PAIR was introduced as a poster presentation at the 24th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval [Wan and Moffat, 2001b], and then subsequently expanded for the 2001 Symposium on String Processing and Information Retrieval [Moffat and Wan, 2001].

Merging compressed blocks, as covered in Chapter 5, was presented at the 2002 Symposium for Combinatorial Pattern Matching [Wan and Moffat, 2002].

The results from the experiments of Chapter 8 on compression with HTML hyperlinks was presented at the 2001 Australasian Database Conference [Wan and Moffat, 2001a].

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